

MadhuTwin

A hybrid digital twin for Type 2 Diabetes that fuses EHR and wearable data to warn doctors up to 2 hours before glucose spikes and lows

MODEL

SYNC

SIMULATE

Team: TODO(team) · College / incubator: TODO(team)

Happiest Health Digital Twin Challenge 2026 · Phase 1

Diabetes care in India is reactive; the risk lives between visits

101 million

adults in India live with diabetes

136 million

more have prediabetes

Source: ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023

A quarterly HbA1c describes the past

It cannot warn about tonight's low on a sulfonylurea or premixed insulin, or tomorrow's illness-driven rise.

Post-meal spikes go unseen

Rice- and roti-heavy meals, late dinners and festival sweets drive excursions that clinic visits never capture.

Data exists, attention does not

CGMs and smartwatches stream thousands of readings a day; no care team can watch hundreds of patients' streams.

MadhuTwin: a living virtual replica of each patient

Not just a predictor: the three defining properties of a digital twin

MODEL

Personalised glucose–insulin physiology

An extended minimal model with meals, exercise, sleep, dawn effect and Indian therapies, fitted to each patient with EHR-based priors.

SYNC

Continuously updated from live data

An Unscented Kalman Filter re-estimates hidden insulin state and a drifting insulin sensitivity every 5 minutes.

SIMULATE

Run forward to predict and explore

Causal 2-hour forecasts and what-ifs: a different meal, a walk after dinner, a short night, an illness.

On top: TwinNet, a physics-gated network that fuses both data streams with the twin's own forecast, and explains every alert in plain language.

Architecture: two data streams fused into the twin

MadhuTwin architecture: two data streams fused into a living digital twin

Type 2 Diabetes · predicts glucose spikes and hypoglycaemia up to 2 hours ahead · doctor-facing dashboard

1 · Data streams

Static / historical: EHR

Demographics, diagnoses (SNOMED CT)
Labs: HbA1c, FPG, insulin, lipids, eGFR (LOINC)
Medications with timing (WHO ATC)
Family history · TCF7L2 rs7903146 · polygenic risk
FHIR R4 bundles, ABDM-aligned

Dynamic / real-time: wearables & IoT

CGM every 5–15 min
Heart rate, HRV (RMSSD), steps, METs
Sleep stages (hypnogram)
Logged meals (Indian foods) and doses

Sources

Synthetic India cohort 1,000 × 14 d · CGMacros (45, real) · ShanghaiT2DM (100, external)

2 · Digital twin core

Ingest & fuse

Common 5-min schema · missing-aware · EHR vector + 21 dynamic channels

MODEL

Personalised glucose–insulin physiology (MAP fit, EHR priors)

SYNC

Unscented Kalman filter every 5 min: state + insulin-sensitivity drift

SIMULATE

2-h forecasts and what-if: meals, walks, sleep, illness

TwinNet: physics-gated fusion network

Causal dilated TCN over 6 h × 21 channels
EHR encoder conditions the dynamics (FiLM)
Gate learns how much to trust the physics twin per horizon
Outputs: 5–120 min quantiles (conformal 80% band) + spike / hypo probability
LightGBM + SHAP: plain-language reasons

3 · Clinician experience

FastAPI + WebSocket

Live replay · audit trail · FHIR export

Doctor dashboard

Risk-ranked remote-monitoring panel
Virtual patient: organ-level physiology
24 h CGM + 2 h forecast cone + alerts with reasons
AGP report (international consensus)
What-if simulator with Indian meals
Ask the twin (grounded assistant)

Interoperability

FHIR R4 RiskAssessment carries the twin's forecast to any EHR

Governance consent artefacts · purpose-tagged audit log · de-identified & synthetic data · model card · DPDP Act 2023 · CDSCO SaMD pathway

Two data streams, three open or synthetic sources

Sandbox rules respected: no real Indian patient data is used

Static / historical: EHR

Demographics · diagnoses (SNOMED CT) · labs incl. HbA1c, fasting glucose & insulin, lipids, eGFR (LOINC) · medications with timing (WHO ATC) · 2-year lab history · family history · TCF7L2 rs7903146 genotype · polygenic risk score

Dynamic / real-time: wearables & IoT

CGM (5- or 15-min) · heart rate · HRV (RMSSD) · steps & METs · sleep stages · logged meals from an Indian food table · insulin and tablet doses

1,000 × 14 days

Synthetic India-calibrated patients

ICMR-INDIAB mix, regional diets, Indian prescribing; streams generated by the same physiology

45 people

CGMacros (PhysioNet)

Real CGM, Fitbit, meals, labs; healthy to T2D (USA)

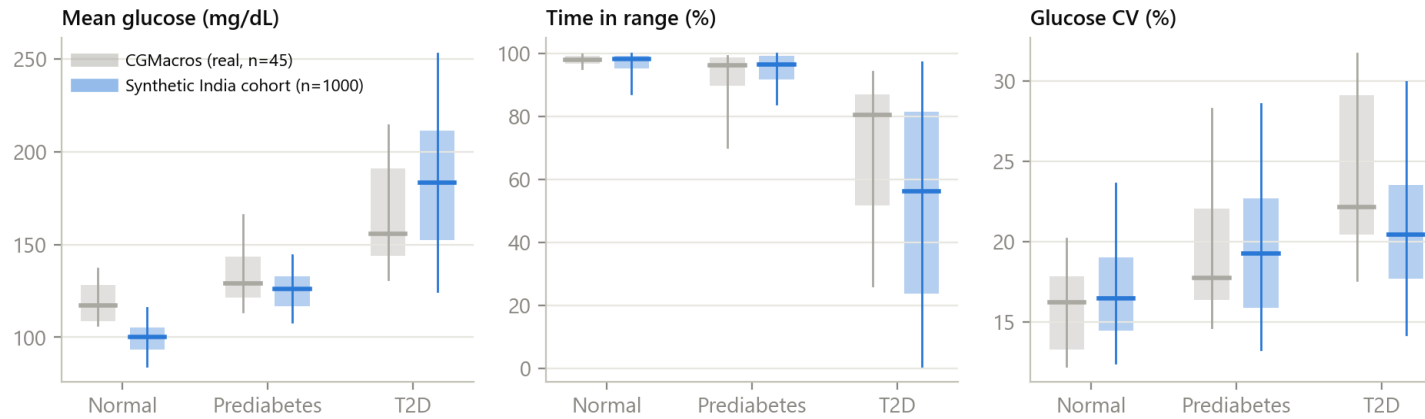
100 patients

ShanghaiT2DM (Figshare)

External: rice-based diet, insulin users, no wearables

A synthetic Indian cohort whose two streams are causally linked

Synthetic CGM statistics vs real people (5th-95th, IQR, median); the Indian T2D cohort is deliberately sicker (median HbA1c 8.0% vs 7.1%)



Physiology parameters are sampled from relationships measured by fitting twins to 45 real people, so synthetic glucose behaves like real glucose.

- Status mix, South-Asian BMI phenotype, comorbidities and prescribing calibrated to Indian clinics: premixed 30/70 insulin, gliclazide, teneligliptin
- Regional meals from IFCT 2017: idli-sambar, roti-sabzi, ragi mudde, biryani, chai with sugar, festival sweets
- Wearables are simulated by the same physiology, and labs are measured from the simulated body (HbA1c from mean glucose)
- Realistic imperfection: unlogged meals, carb misestimation, CGM noise and compression lows, missed doses, illness and stress days

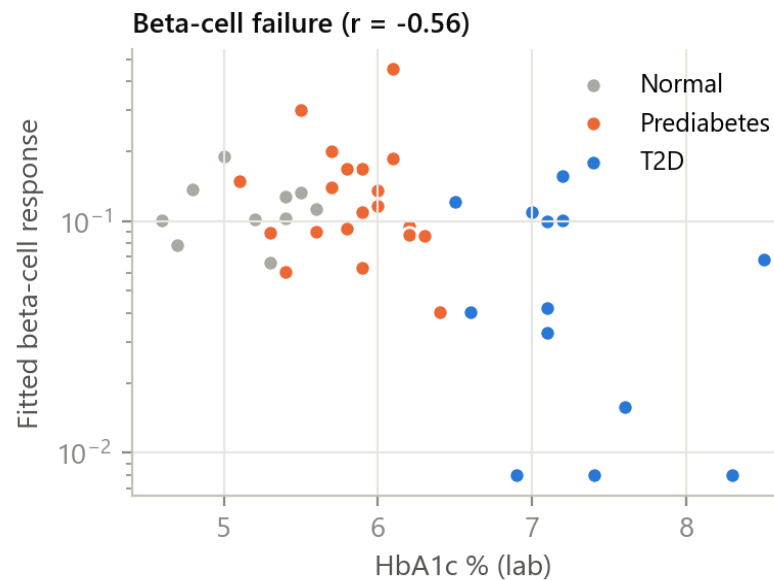
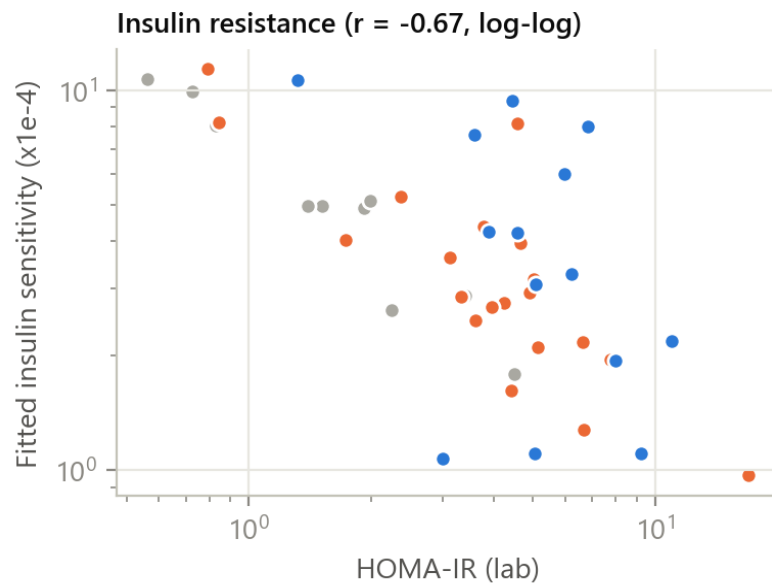
MODEL: personal twins recover known physiology from glucose data

MODEL

SYNC

SIM

Twins fitted to 45 real people recover known physiology from CGM



37.7 → 26.6

mg/dL twin error, EHR prior → personalised (45 real people, 1–4 h windows)

29% lower

error once the twin is fitted to the individual

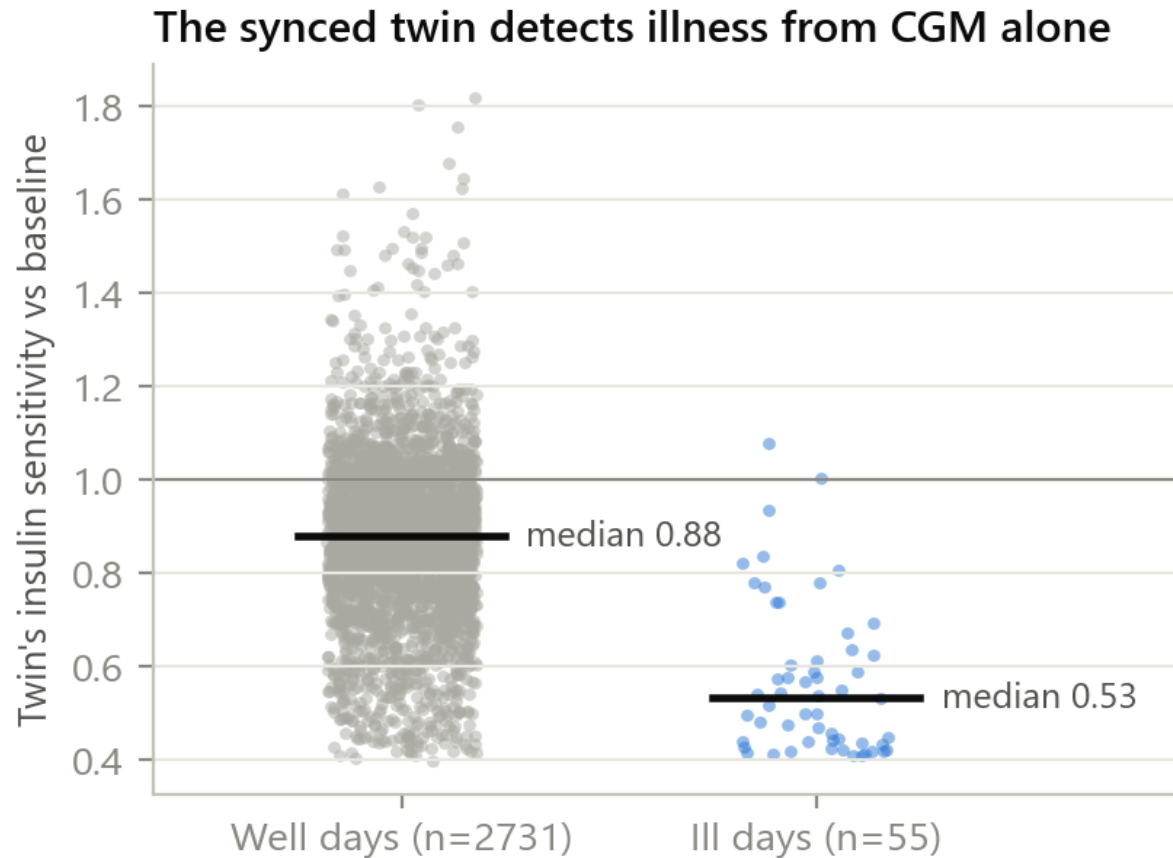
Fitted insulin sensitivity tracks lab HOMA-IR and beta-cell response tracks HbA1c, without the model ever seeing those relationships during fitting.

SYNC: the twin notices illness from CGM alone

MODEL

SYNC

SIM



0.89

AUROC for identifying ill days from the synced insulin-sensitivity trace (55 ill days, 2786 patient-days)

0.57x vs 0.87x

insulin sensitivity relative to baseline on ill vs well days

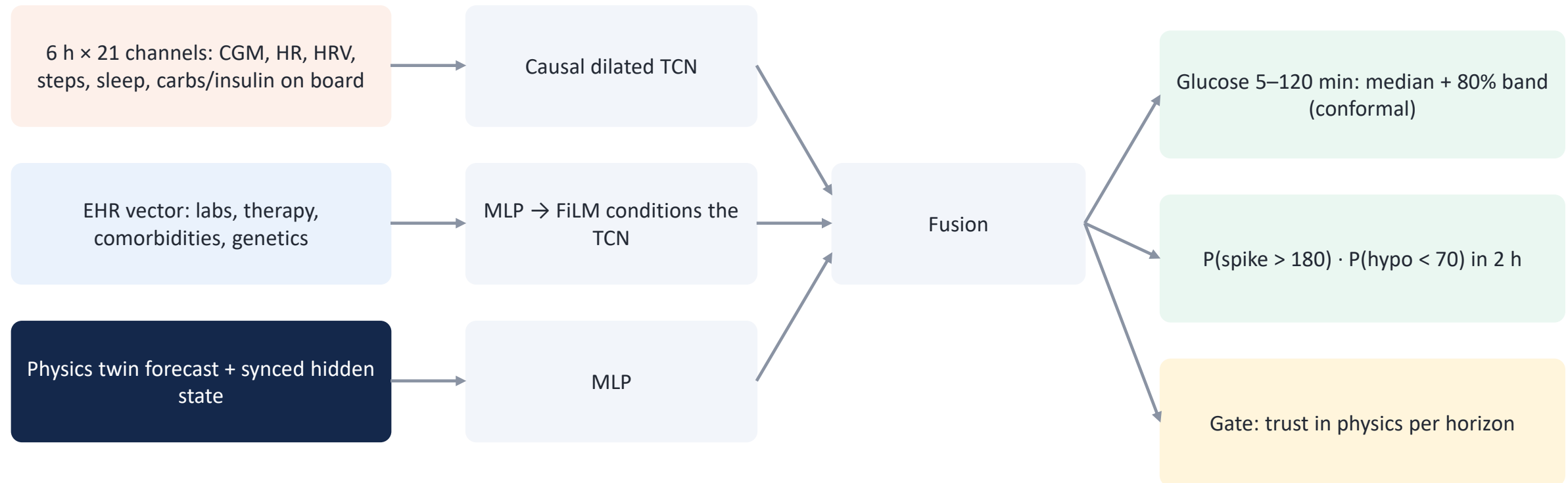
The Unscented Kalman Filter tracks a drifting insulin-sensitivity factor every 5 minutes. A sustained drop prompts the dashboard insight: check for infection, stress or steroid use.

SIMULATE + learn: TwinNet fuses both streams with the twin

MODEL

SYNC

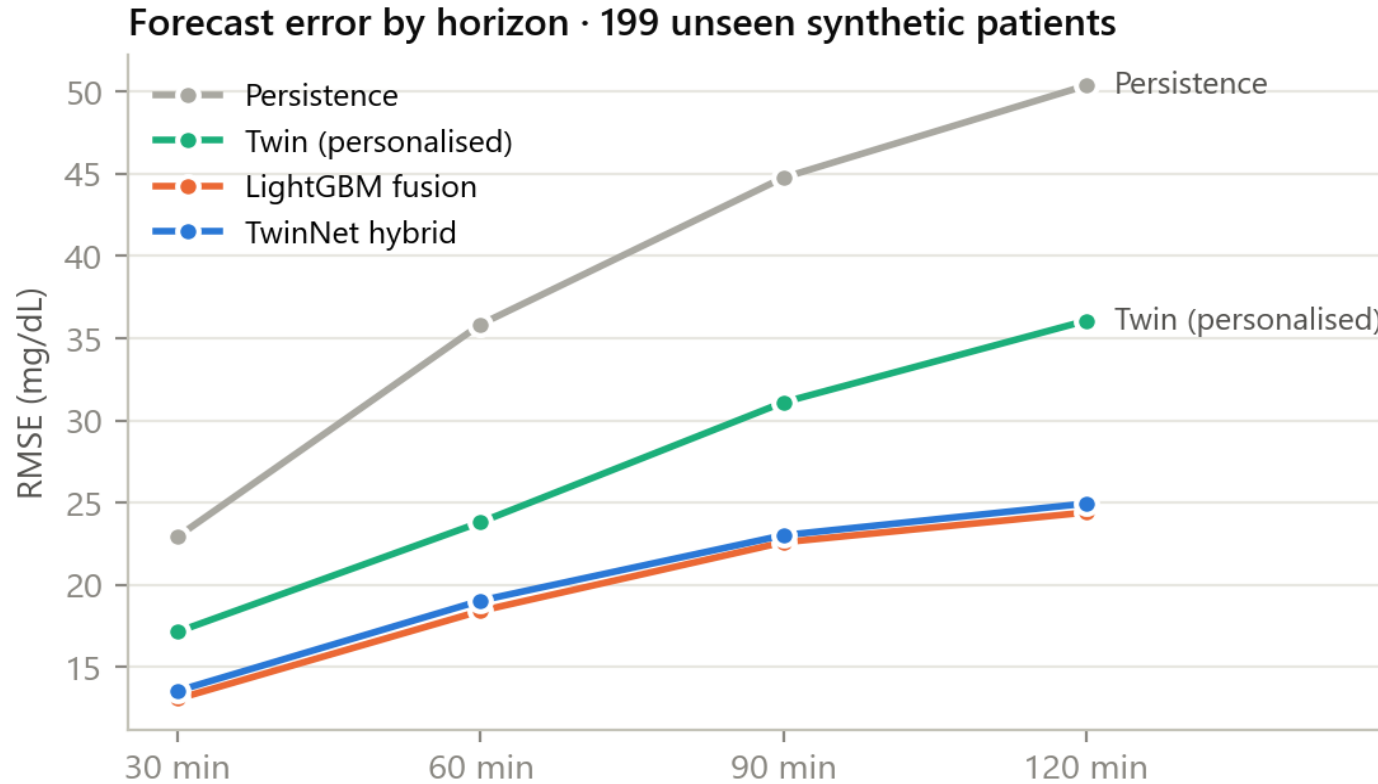
SIM



Learned trust in the physics twin rises with horizon: 0.71 at 30 min, 0.82 at 60 min, 0.92 at 2 h. Modality dropout during training lets the same model run without wearables or EHR. Pretrained on synthetic patients, fine-tuned on real ones.

Accurate 2-hour glucose forecasts on unseen patients

199 held-out patients · days 8–14 after personalisation · 1,29,956 forecasts



19.0 mg/dL

60-min error (RMSE); persistence 35.8 · LightGBM 18.4

99.4%

of 60-min forecasts in Clarke zones A+B (clinically acceptable)

80%

of outcomes inside the 80% interval (target 80%)

Early warnings with lead time a clinician can act on

86%

of sustained spikes (>180) flagged in advance

100 min

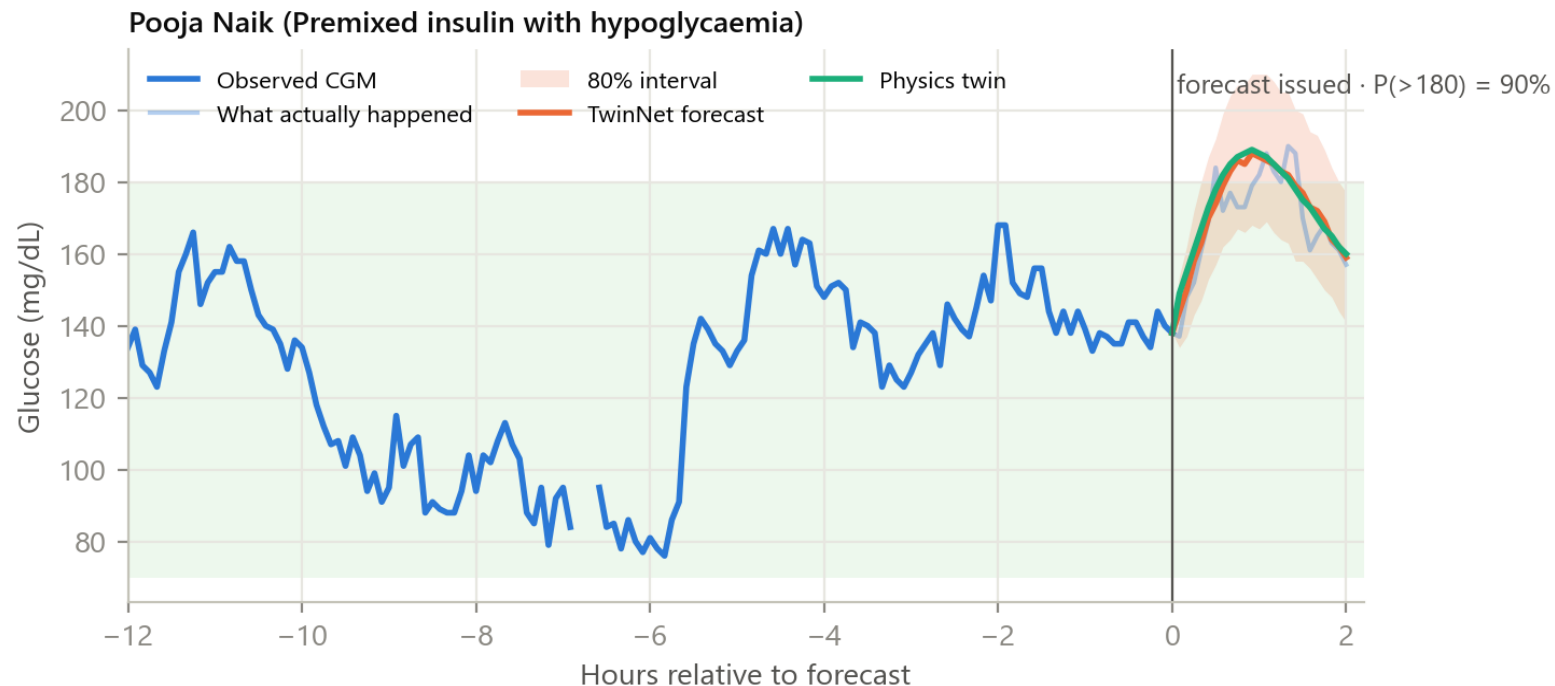
median warning before the spike begins

0.45

false alert episodes per patient-day

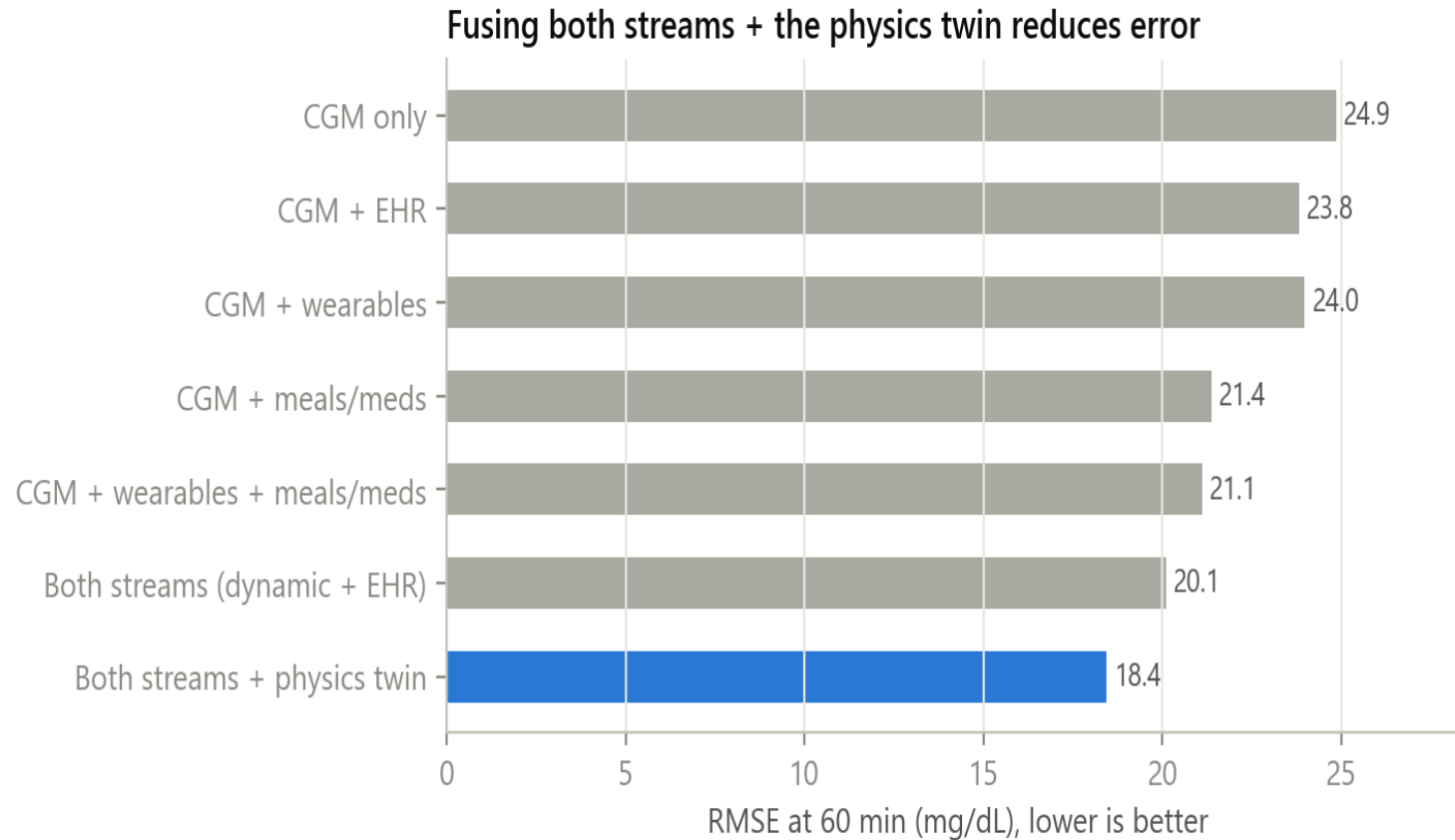
0.97

hypoglycaemia AUROC (spike AUROC 0.96)



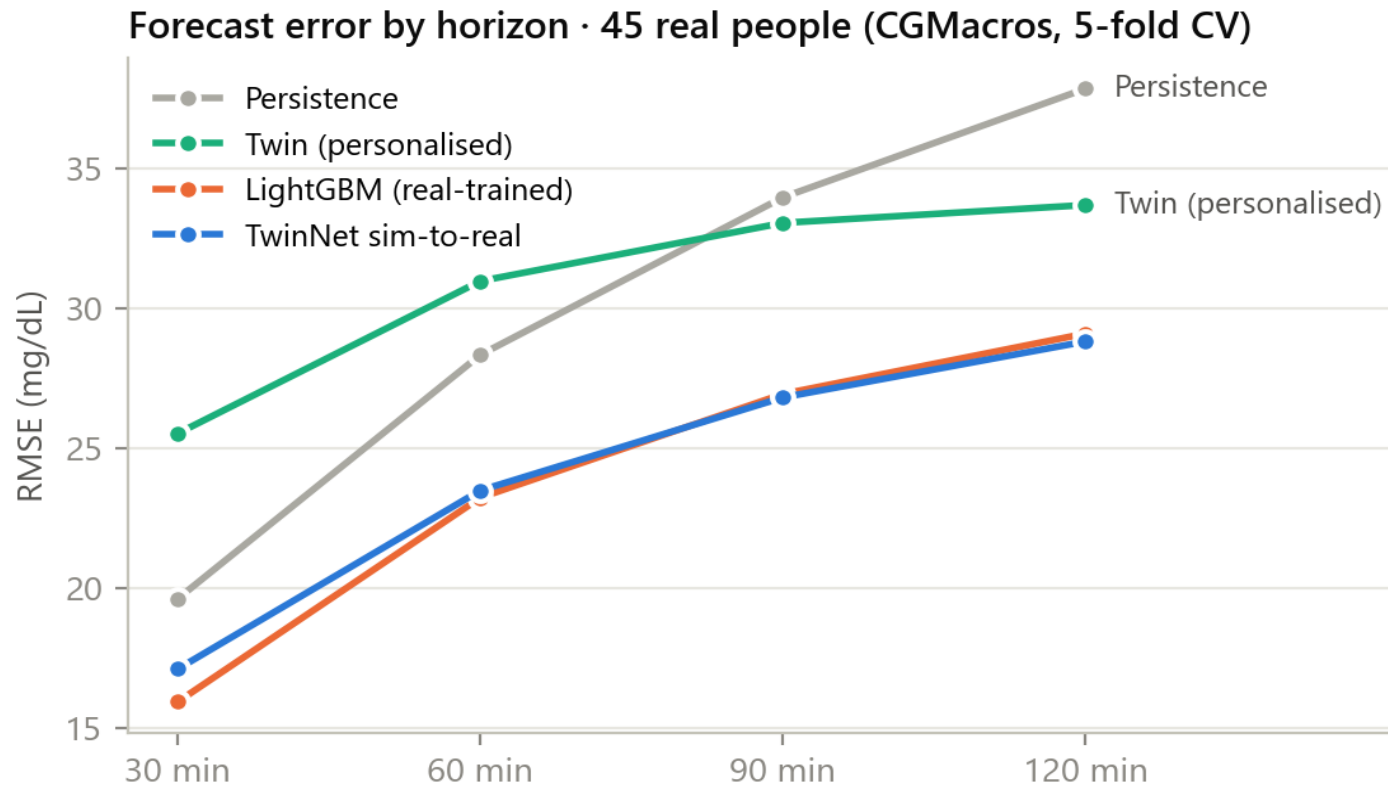
Every alert carries its reasons from the event model, e.g. carbohydrate still absorbing, a rising trend, insulin sensitivity below baseline, or a sedentary hour after a meal.

Fusing both streams matters, and the twin adds the most



- CGM alone: 24.9 mg/dL at 60 min; spike AUROC 0.92
- Both streams (wearables + meals/meds + EHR): 20.1; AUROC 0.95
- + physics twin: 18.4; AUROC 0.96
- Each stream helps; the EHR helps most in combination, telling the model whose body the wearables describe

It transfers to real people and to an external population



60-min RMSE (mg/dL)	CGMacros	Shanghai
Persistence	28.3	27.5
TwinNet, synthetic only	26.4	32.9
TwinNet, sim-to-real	23.5	22.3
Clarke A+B, sim-to-real	99.4%	98.9%

45 CGMacros participants (5-fold patient cross-validation) and 109 ShanghaiT2DM recordings from a different country, sensor and diet, with no wearables at all.

CGM-light: one week of CGM, then fingersticks and a smartwatch

An affordability mode for India: the twin keeps estimating continuous glucose after the sensor comes off

6.4%

twin MARD vs true glucose (carry-forward fingersticks: 23.8%)

95%

of twin estimates within 20% of true glucose (carry-forward: 56%)

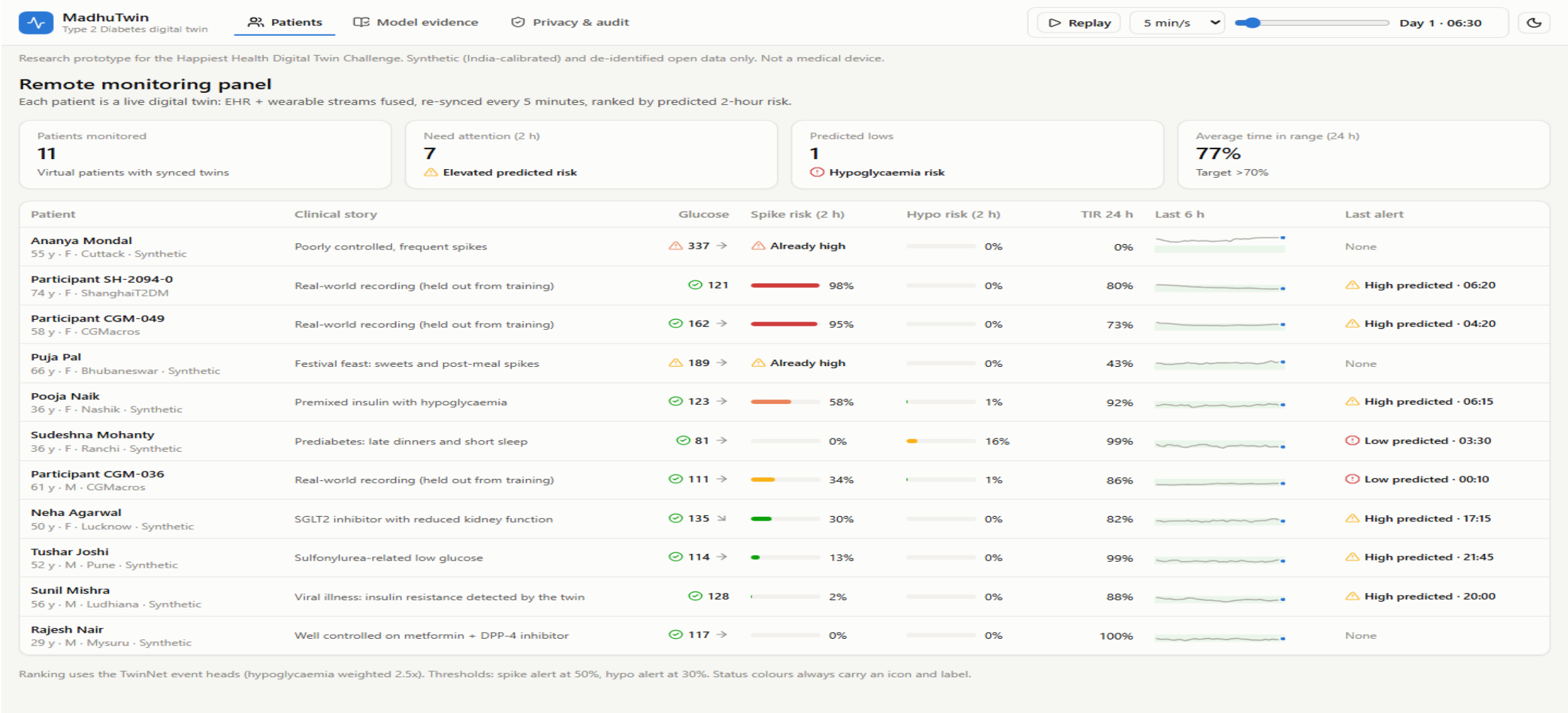
± 3.8 pp

error in weekly time-in-range (fingersticks alone: ± 5.4 pp)

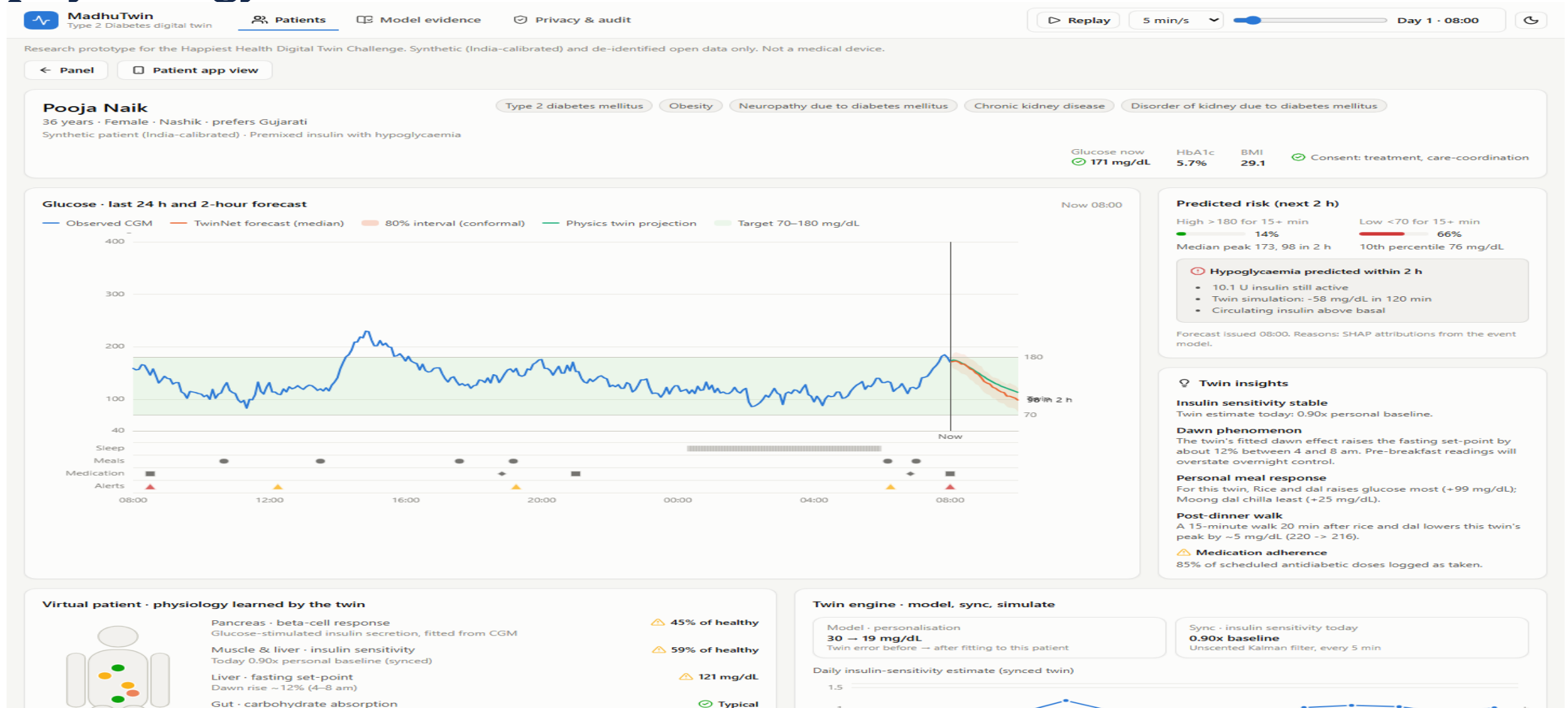
How it works

Week 1 personalises the twin on CGM. In week 2 only 4 fingerstick checks a day remain; the Unscented Kalman Filter fuses them with logged meals, doses and smartwatch activity to estimate glucose in between. Evaluated on 199 held-out synthetic patients against their true glucose; because the simulator shares the twin's structure this is an upper bound until tested on real CGM-off data.

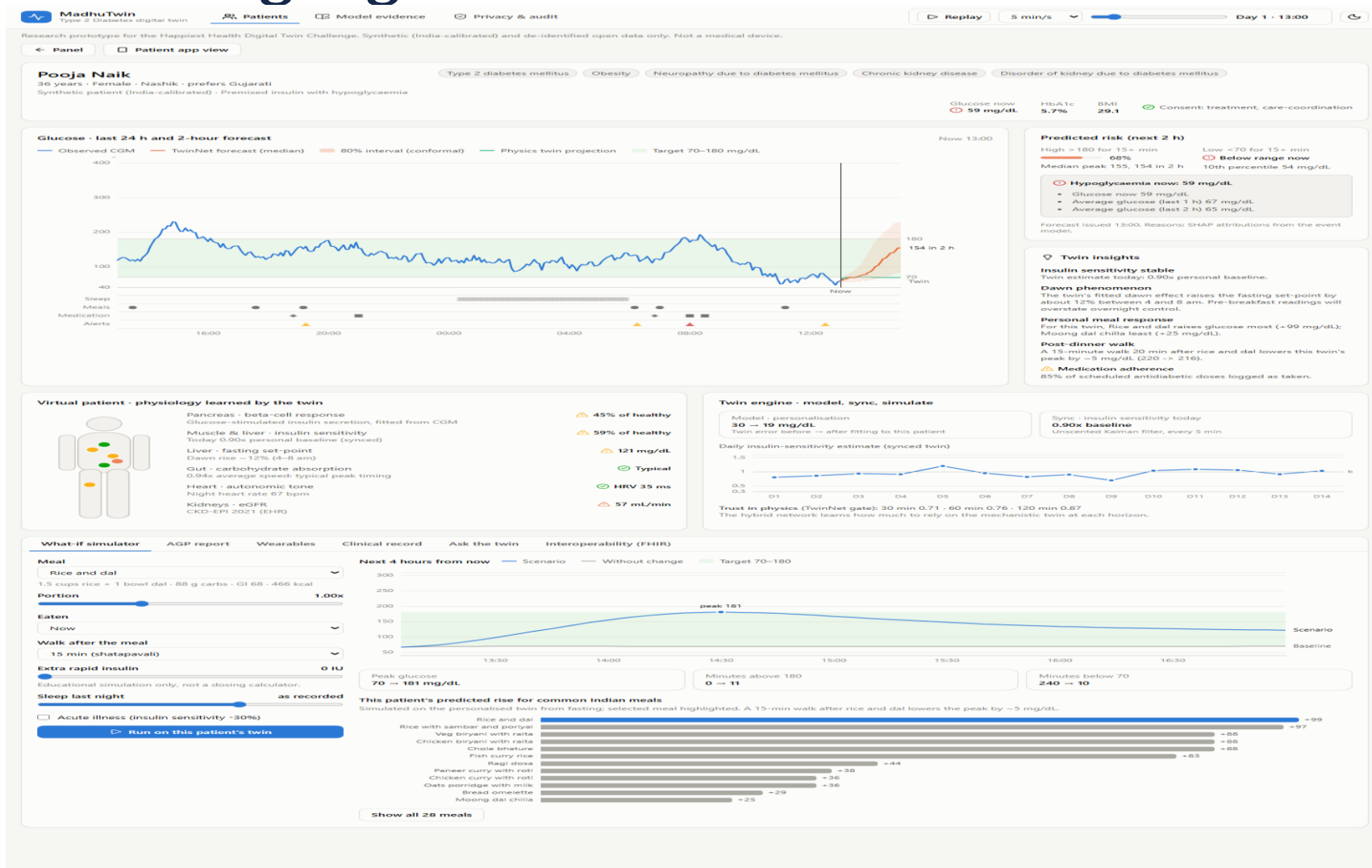
The doctor's view: a risk-ranked remote-monitoring panel



The virtual patient: forecast, reasons and learned physiology



What-if on the patient's own twin, and a companion in their language



Responsible by design: privacy, interoperability, explainability

1

DPDP Act 2023

Consent artefacts, purpose limitation, data minimisation, purpose-tagged audit trail; synthetic and de-identified data only

2

ABDM-ready interoperability

FHIR R4 with SNOMED CT, LOINC, WHO ATC; CGM as SampledData; the forecast exported as a RiskAssessment

3

Explainable

Reasons on every alert (SHAP), a visible trust-in-physics gate, organ-level physiology, AGP report

4

Honest evaluation

Patient-level splits, external validation, bootstrap CIs, model card with limitations

5

Regulatory path

Decision support, not autonomous treatment: CDSCO Software as a Medical Device route with prospective validation

From proof of concept to Indian clinics



Now

Proof of concept

Twin, TwinNet, dashboard, FHIR;
validated on synthetic, US and Chinese
data



3 months

Co-design

Diabetologist review of alerts and
thresholds; native-speaker review of
patient text



6 months

Pilot

50–100 people with T2D in Bengaluru
clinics: CGM + smartwatch + ABDM-
linked EHR



12 months

Validation

Prospective accuracy and outcome
study; CDSCO SaMD submission

Impact we are aiming for

Fewer nocturnal hypoglycaemia episodes in people on sulfonylureas and premixed insulin, more time in range through personalised meal and activity advice, and a care team that sees risk before it becomes an emergency.

Thank you

MadhuTwin: model · sync · simulate, for every person living with diabetes

Code, data pipeline, evaluation and dashboard: GitHub repository [TODO\(team\)](#) · Apache-2.0

Team: [TODO\(team\)](#)